WORKED SOLUTION

STEP III 2025, Question 6

Pure • Symmetric Functions of Roots

STEP III 2025, Question 6

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

The key idea

Conditions on $a+b+c$, $a^2+b^2+c^2$ and $a^3+b^3+c^3$ are really conditions on the elementary symmetric functions $a+b+c$, $ab+bc+ca$ and abc. Once you translate, the numbers become the roots of a very simple polynomial (in (i)), or a classic factorisation does all the work (in (ii) and (iii)). The final part is the same algebra run in reverse.

Solution

Part (i)

Squaring $a+b+c=0$ gives

$$0=(a+b+c)^2=a^2+b^2+c^2+2(ab+bc+ca)=2(ab+bc+ca),$$

so $ab+bc+ca=0$ as well.

Not all real. If a, b and c were all real, then a^2, b^2, c^2 would be non-negative reals with sum 0, forcing $a=b=c=0$. This contradicts the numbers being non-zero, so they cannot all be real.

Equal moduli. Write $p=abc$, which is non-zero. Since $a+b+c=0$ and $ab+bc+ca=0$,

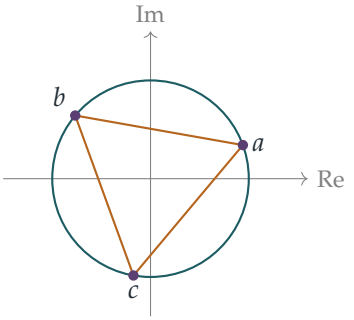
$$(z-a)(z-b)(z-c)=z^3-(a+b+c)z^2+(ab+bc+ca)z-abc=z^3-p.$$

So each of a, b, c is a root of $z^3=p$, that is,

$$a^3=b^3=c^3=p.$$

Taking moduli, $|a|^3=|b|^3=|c|^3=|p|$, and since moduli are non-negative real numbers, $|a|=|b|=|c|=|p|^{1/3}$. So a, b and c all have the same modulus, as required.

In fact a, b, c are the three cube roots of p, which sit at the vertices of an equilateral triangle centred on the origin.



Part (ii)

Suppose $a+b+c=0$. Then $c=-(a+b)$ and

$$a^3+b^3+c^3 = a^3+b^3-(a+b)^3 = -3a^2b-3ab^2 = -3ab(a+b) = 3abc.$$

If also $a^3+b^3+c^3=0$, then $abc=0$, so at least one of the numbers is zero. Hence there are no three *non-zero* complex numbers with both properties.

Part (iii)

The useful identity this time is

$$(a+b+c)^3 = a^3+b^3+c^3+3(a+b)(b+c)(c+a),$$

which you can verify by expanding: both sides contain each of a^2b, ab^2 and so on three times, and abc six times.

Now $d=-(a+b+c)$, so $d^3=-(a+b+c)^3$ and

$$0 = a^3+b^3+c^3+d^3 = a^3+b^3+c^3-(a+b+c)^3 = -3(a+b)(b+c)(c+a).$$

Ask yourself Why does a product being zero tell us anything about moduli?

A product of complex numbers is zero only if one factor is zero. So one of $a+b$, $b+c$, $c+a$ vanishes; say $a+b=0$ (the other cases are the same with letters relabelled). Then $b=-a$, so $|b|=|a|$. Hence at least two of the four numbers have the same modulus.

(Nothing here used $d\neq 0$ or that the numbers are non-zero; the conclusion is simply that the four numbers split into two pairs of negatives, as in $a,-a,c,-c$.)

Part (iv)

With $c=1$, $d=-2$, $e=3$ we have

$$c+d+e=2, \qquad c^3+d^3+e^3 = 1-8+27 = 20,$$

so we need real a, b with

$$a+b=-2, \qquad a^3+b^3=-20.$$

Using $a^3+b^3=(a+b)^3-3ab(a+b)$, the second condition becomes

$$-20=-8+6ab \quad\Longrightarrow\quad ab=-2.$$

So a and b are the roots of $t^2+2t-2=0$, namely

$$a=-1+\sqrt{3}, \qquad b=-1-\sqrt{3},$$

which are real since the discriminant $4+8=12$ is positive. Their magnitudes are $\sqrt{3}-1\approx 0.73$ and $\sqrt{3}+1\approx 2.73$, neither of which is 1, 2 or 3 (both are irrational). So

$$-1+\sqrt{3}, \quad -1-\sqrt{3}, \quad 1, \quad -2, \quad 3$$

are five real numbers with distinct magnitudes, summing to 0, whose cubes also sum to 0.

Checks

- In (i), the cube roots of 1, namely 1, ω, ω^2, satisfy $1+\omega+\omega^2=0$ and $1+\omega^2+\omega^4=1+\omega^2+\omega=0$: all modulus 1, not all real, as predicted.
- In (iii), try $a=2$, $b=-2$, $c=\mathrm{i}$, $d=-\mathrm{i}$: both sums vanish, and indeed the numbers pair off into negatives.
- In (iv), numerically $0.7320-2.7320+1-2+3=0$ and $0.3923-20.3923+1-8+27=0$, using $(\sqrt{3}-1)^3=6\sqrt{3}-10$ and $(-1-\sqrt{3})^3=-6\sqrt{3}-10$.

Common slips

- In (i), concluding $|a|=|b|$ from $a^3=b^3$ by "taking cube roots": a and b need not be equal, but their moduli are, because moduli are non-negative reals.
- In (iii), forgetting to say *why* $(a+b)(b+c)(c+a)=0$ forces one factor to be zero (there are no zero divisors in $\mathbb{C}$).
- In (iv), not checking that the magnitudes of a and b differ from 1, 2, 3 and from each other.

Beyond STEP

Every symmetric polynomial in several variables can be written in terms of the elementary symmetric polynomials, and Newton's identities give the link with the power sums $a^k+b^k+c^k+\dots$ explicitly; this is why the conditions here pin the numbers down so tightly. A close cousin is the Prouhet–Tarry–Escott problem: find two different sets of integers whose sums of first, second, …, kth powers all agree. It is still partly open. For the algebra of polynomials and their roots, see Allenby, *Rings, Fields and Groups*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.